Nightingale: reading list

Ten papers behind the project's data, clinical rules, methods and explanation layer · compiled 2026-11-07

> Suggested order: papers 1 to 3 give the setting; 4 and 5 are short and explain two of our rules; 6 to 8 explain the methods we use and measure; 9 and 10 cover the explanation layer. About one to two hours for the whole set if you read only the parts named under each paper.

The problem and the data

1

DDXPlus: A New Dataset For Automatic Medical Diagnosis

Fansi Tchango A, Goel R, Wen Z, Martel J, Ghosn J
NeurIPS 2022, Datasets and Benchmarks track. arXiv:2205.09148

arXiv: arxiv.org/abs/2205.09148

Open (arXiv)

Why it matters here. Our only patient data. It explains how the synthetic patients are generated from a medical knowledge base, which is exactly why a model scores near 100% on it and why the graph score is circular (risks R-12 and R-16).

Read. The dataset construction section, then how the differential is built. Skim the baselines.

2

An overview of clinical decision support systems: benefits, risks, and strategies for success

Sutton RT, Pincock D, Baumgart DC, Sadowski DC, Fedorak RN, Kroeker KI
npj Digital Medicine 3, 17 (2020). doi:10.1038/s41746-020-0221-y

DOI: doi.org/10.1038/s41746-020-0221-y

Open (CC BY 4.0)

Why it matters here. The case for decision support that informs rather than decides, and the risks that come with it, alert fatigue among them. It frames why we show a ranked list with reasons and measure the share of needless red flags.

Read. The sections on risks and on strategies for success.

The clinical content

3

2021 AHA/ACC/ASE/CHEST/SAEM/SCCT/SCMR Guideline for the Evaluation and Diagnosis of Chest Pain

Gulati M, Levy PD, Mukherjee D, Amsterdam E, Bhatt DL, Birtcher KK, et al.
Circulation 2021;144(22). doi:10.1161/CIR.0000000000001029

DOI: doi.org/10.1161/CIR.0000000000001029

Journal page (access may need your university login)

Why it matters here. The clinical reference for acute chest pain: which causes must not be missed, and which investigations come first (ECG, troponin). It is the background for our must-not-miss list and for the tests the dashboard suggests.

Read. The summary tables and the sections on acute chest pain in the emergency setting. It is long; do not read it end to end.

4

The International Registry of Acute Aortic Dissection (IRAD): New Insights Into an Old Disease

Hagan PG, Nienaber CA, Isselbacher EM, Bruckman D, Karavite DJ, Russman PL, et al.
JAMA 2000;283(7):897. doi:10.1001/jama.283.7.897

DOI: doi.org/10.1001/jama.283.7.897

Journal page (access may need your university login)

Why it matters here. Where the frequencies of aortic dissection's presenting findings come from. Our hand-written dissection edges in the knowledge graph are weighted from IRAD figures, since DDXPlus has no dissection patients.

Read. The tables of presenting symptoms and signs.

5

Sensitivity of the Aortic Dissection Detection Risk Score, a Novel Guideline-Based Tool for Identification of Acute Aortic Dissection at Initial Presentation

Rogers AM, Hermann LK, Booher AM, Nienaber CA, Williams DM, Kazerooni EA, et al.
Circulation 2011;123(20):2213-2218. doi:10.1161/CIRCULATIONAHA.110.988568

DOI: doi.org/10.1161/circulationaha.110.988568

Journal page (access may need your university login)

Why it matters here. The ADD-RS, which our aortic-dissection red-flag rule follows. It shows how a red flag can be built from groups of findings instead of a single symptom, the reason our rule no longer fires on back pain alone.

Read. The whole paper; it is short.

The methods

6

Learning a Health Knowledge Graph from Electronic Medical Records

Rotmensch M, Halpern Y, Tlimat A, Horng S, Sontag D
Scientific Reports 7, 5994 (2017). doi:10.1038/s41598-017-05778-z

DOI: doi.org/10.1038/s41598-017-05778-z

Open (CC BY 4.0)

Why it matters here. A knowledge graph linking symptoms to diseases, built from real hospital records and used to rank diagnoses with simple probabilistic models. A useful contrast with ours, which is built from the DDXPlus definitions and so inherits the circularity risk.

Read. How the graph is learned and how its diagnostic ranking is evaluated.

7

Revisiting Deep Learning Models for Tabular Data

Gorishniy Y, Rubachev I, Khrulkov V, Babenko A
NeurIPS 2021. arXiv:2106.11959

arXiv: arxiv.org/abs/2106.11959

Open (arXiv)

Why it matters here. The source of the transformer design we adopted (decision D-12): one token per feature, a small stack of attention blocks, default settings fixed in advance. It also shows that gradient boosting and a plain ResNet stay strong on tabular data, so a deep model has to be compared with simple baselines, as we do with logistic regression and XGBoost.

Read. The FT-Transformer section and the comparison with gradient boosting.

8

On Calibration of Modern Neural Networks

Guo C, Pleiss G, Sun Y, Weinberger KQ
ICML 2017. arXiv:1706.04599

arXiv: arxiv.org/abs/1706.04599

Open (arXiv)

Why it matters here. Defines the expected calibration error (ECE) and the reliability diagram we report, and compares temperature scaling with Platt scaling and isotonic regression, the methods in our calibration experiment (EXP-007).

Read. The definitions of ECE and the reliability diagram, then the results on temperature scaling.

Explanation and grounding

9

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Lewis P, Perez E, Piktus A, Petroni F, Karpukhin V, Goyal N, et al.
NeurIPS 2020. arXiv:2005.11401

arXiv: arxiv.org/abs/2005.11401

Open (arXiv)

Why it matters here. The idea behind our evidence layer and the grounded explainer: retrieve passages first, then let the language model write from them. Our version retrieves for explanation only and never uses the passages to rank.

Read. The overview figure and the method section.

10

Survey of Hallucination in Natural Language Generation

Ji Z, Lee N, Frieske R, Yu T, Su D, Xu Y, et al.
ACM Computing Surveys 55 (2023), 1-38. doi:10.1145/3571730. arXiv:2202.03629

arXiv (open): arxiv.org/abs/2202.03629 · DOI: doi.org/10.1145/3571730

Open (arXiv version)

Why it matters here. Why a language model's explanation can sound right and be unsupported, and how that is measured. It is the background for our guardrails, which check each sentence against what the result holds, and for the unsupported-claim rate in the evaluation.

Read. The definitions of intrinsic and extrinsic hallucination and the section on metrics.

Titles, authors, years and venues were checked against Crossref (journal articles) and arXiv (preprints) on 2026-11-07. Where a page asks for a login, your university library usually gives access. Volume and page details are as the publisher records them; confirm them against the publisher's page before putting a reference in the report.